

CIV102F - Structures and Materials Equations

Miles Ehrlich

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1 Basic Concepts

1.1 Equilibrium

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum M_P = 0 \quad (1)$$

$$F_y = F_x \tan(\theta) \quad (2)$$

1.2 Moment

$$M = F \cdot d_{\perp} \quad (3)$$

1.3 Stress-Strain

$$\sigma = \frac{F}{A} \quad (4)$$

$$\varepsilon = \frac{\Delta}{L_0} \quad (5)$$

$$\sigma = E \cdot \varepsilon = k \frac{L_0}{A} \cdot \varepsilon \quad (6)$$

$$\Delta = \frac{FL_0}{EA} \quad (7)$$

$$U = \int \sigma d\varepsilon = \frac{1}{2} \sigma \varepsilon \quad (8)$$

$$W = U \cdot V_0 \quad (9)$$

$$R = \int_0^{\varepsilon_y} \sigma d\varepsilon \quad (10)$$

$$T = \int_0^{\varepsilon_r} \sigma d\varepsilon \quad (11)$$

2 Suspension Bridges

$$A_y = B_y = \frac{wL}{2} \quad (12)$$

$$H = \frac{wL^2}{8h} \quad (13)$$

$$T_{max} = \sqrt{H^2 + A_y^2} \quad (14)$$

3 Structural Analysis

3.1 Pulleys

$$T_1 = T_2 \quad (15)$$

3.2 Rollers

$$\sum F_x = 0 \quad (16)$$

3.3 Pins, Hinges

$$\sum F_x = \sum F_y = 0 \quad (17)$$

3.4 Fixed End

$$\sum F_x = \sum F_y = \sum M = 0 \quad (18)$$

4 Oscillations

4.1 Simple Harmonic Motion

$$F = -kx \quad (19)$$

$$x(t) = A \sin(\omega t) + \Delta_0 \quad (20)$$

$$\omega = \sqrt{\frac{k}{m}} \quad (21)$$

$$f_{n,point} = \frac{15.76}{\sqrt{\Delta_0}} \quad (22)$$

$$f_{n,UDL} = \frac{17.76}{\sqrt{\Delta_0}} \quad (23)$$

4.2 Driven Damped Oscillation

$$F(t) = F_0 \sin(\omega t) \quad (24)$$

$$x(t) = DAF \cdot \frac{F_0}{k} \sin(\omega t + \phi) + \Delta_0 \quad (25)$$

$$DAF = \frac{1}{\sqrt{\left(1 - \left(\frac{f}{f_n}\right)^2\right)^2 + \left(\frac{2\beta f}{f_n}\right)^2}} \quad (26)$$

$$F_{max} = F_{stat} + DAF \cdot F_0 \quad (27)$$

$$\Delta x_{max} = \Delta x_{stat} \cdot \frac{F_{max}}{F_0} \quad (28)$$

5 Cross-Sections

$$\bar{y} = \frac{\sum A_i y_i}{A_i} \quad (29)$$

$$I = \int_M y^2 dA \quad (30)$$

$$I = \sum I_0 + A_i d_i^2 \quad | \quad d = \bar{y} - y_i \quad (31)$$

$$I_{Rect} = \frac{bh^3}{12} \quad (32)$$

$$I_{Circ} = \frac{\pi^2 d^4}{64} \quad (33)$$

$$Q = \int_{y_{\tau=0}}^y y dA \quad (34)$$

$$Q = \sum A_i d_i \quad (35)$$

6 Bending

$$\phi = \frac{d\theta}{dx} = \frac{d^2 y}{dx^2} \quad (36)$$

$$\varepsilon = \phi y \quad (37)$$

$$M = EI\phi \quad (38)$$

7 Euler Buckling

$$P_{crit} = \frac{n^2 \pi^2 EI}{L^2} \quad (39)$$

$$r = \sqrt{\frac{I}{A}} \quad (40)$$

$$\sigma_{crit} = \frac{\pi^2 E}{\left(\frac{L}{r}\right)^2} \quad (41)$$

$$\Delta_{lat} = \frac{\Delta_0}{1 - \frac{P}{P_{crit}}} \quad (42)$$

8 Trusses

$$A > \frac{2F_{max}}{\sigma_y} \quad (43)$$

$$I > \frac{3L^2 C_{max}}{\pi^2 E} \quad (44)$$

$$r > \frac{L}{200} \quad (45)$$

$$w_{wind} = 2 \text{ kPa} \quad (46)$$

$$A_{wind} = \sum \frac{L_i}{2} \quad (47)$$

$$A_{wind, rail} = h \cdot s \quad (48)$$

$$\Delta_{brace} = 0.01L \quad (49)$$

$$R_{brace} = 0.02P \quad (50)$$

$$P^* \cdot \Delta_B = \sum P_i^* \cdot \frac{P_i L_0}{EA} \quad (51)$$

9 Beams

9.1 Stress Resultants

$$\sigma = \frac{My}{I} + \frac{N}{A} \quad (52)$$

$$\tau_c = \frac{VQ_{max}}{Ib_c} \quad (53)$$

$$\tau_g = \frac{VQ_g}{Ib_g} \quad (54)$$

$$V(x) = \int w(x) dx + P_0 \quad (55)$$

$$M(x) = \int V(x) dx + M_0 \quad (56)$$

$$\phi(x) = \frac{M(x)}{EI} \quad (57)$$

$$E_{w, design} = \frac{E_{05}}{1.5} \quad (58)$$

$$E_{w, deflection} = E_{50} \quad (59)$$

9.2 Deflections/Moment Area Theorems

$$\Delta\theta_A^B = \int_A^B \phi(x) dx \quad (60)$$

$$\delta_{BA} = \int_A^B x \phi(x) dx = \sum \bar{x}_i \cdot A_{\phi, i} \quad (61)$$

9.3 Local Buckling

$$\mu = -\frac{\varepsilon_y}{\varepsilon_x} \quad (62)$$

$$E_{apparent} = \frac{E}{1 - \mu^2} \quad (63)$$

$$\sigma_1 = 4 \cdot \frac{\pi^2 E}{12(1 - \mu^2)} \cdot \left(\frac{t}{b}\right)^2 \quad (64)$$

$$\sigma_2 = 0.425 \cdot \frac{\pi^2 E}{12(1 - \mu^2)} \cdot \left(\frac{t}{b}\right)^2 \quad (65)$$

$$\sigma_3 = 6 \cdot \frac{\pi^2 E}{12(1 - \mu^2)} \cdot \left(\frac{t}{b}\right)^2 \quad (66)$$

$$\tau = 5 \cdot \frac{\pi^2 E}{12(1 - \mu^2)} \cdot \left(\left(\frac{t}{b}\right)^2 + \left(\frac{t}{h}\right)^2 \right) \quad (67)$$

10 Concrete

10.1 Properties

$$f'_t = 0.33 \sqrt{f'_c} \quad (68)$$

$$E_c = 4730 \sqrt{f'_c} \quad | \quad f_c < 0.4 f'_c \quad (69)$$

10.2 Flexure

$$T = A_s \varepsilon E_s \quad (70)$$

$$\eta = \frac{E_s}{E_c} \quad (71)$$

$$\rho = \frac{A_s}{bd} \quad (72)$$

$$K = \sqrt{(\eta\rho)^2 + 2\eta\rho} - \eta\rho \quad (73)$$

$$j = (1 - \frac{K}{3}) \quad (74)$$

$$f_s = \frac{M}{A_s j d} \quad (75)$$

$$f_c = \frac{K}{1 - K} \frac{M}{\eta A_s j d} \quad (76)$$

10.3 Shear

$$v = \frac{V}{b_w j d} \quad (77)$$

$$V_c + V_s = V_r \quad (78)$$

$$V < 0.5V_c + 0.6V_s < 0.125b_w j d \cdot f'_c \quad (79)$$

$$V_{c,weak} = \frac{230\sqrt{f'_c}}{1000 + 0.9d} b_w j d \quad (80)$$

$$\frac{A_s f_y}{b_w s} \geq 0.06 f'_c \quad (81)$$

$$V_{c,strong} = 0.18\sqrt{f'_c} b_w j d \quad (82)$$

$$V_s = \frac{A_v f_y j d \cot(35^\circ)}{s} \quad (83)$$

$$s < 600\text{mm} \quad (84)$$

$$s < 0.63d \quad (85)$$